

Reservoir Characterisation and Modelling Introduction for CSMP Reservoir Simulator

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Model building goals

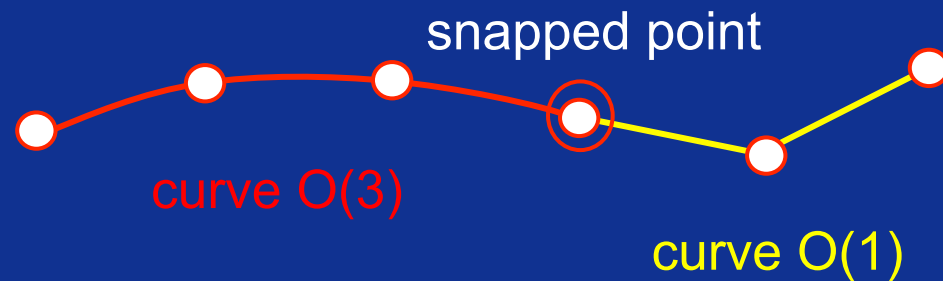
1. Compilation of field data into three-dimensional reservoir model
2. Conversion of visually correct geometric model into water-tight CAD
3. Meshing of CAD geometry
 - a. Meshing with surface-type fracture representations
 - b. Meshing with fractures represented by multiple layers of prism elements
 - c. Conversion of triangle-tetrahedral meshes into hybrid element meshes
 - d. Mesh smoothing and optimisation
 - e. Output of mesh to CSP reservoir simulator
4. Alternative meshing workflow using faceted model geometry as generated by geological modelling / seismic interpretation tools

1. Learn about CAD model building and where specific geometry construction rules come from

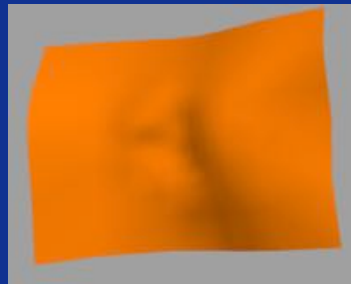
What are models build from ?

- Points 

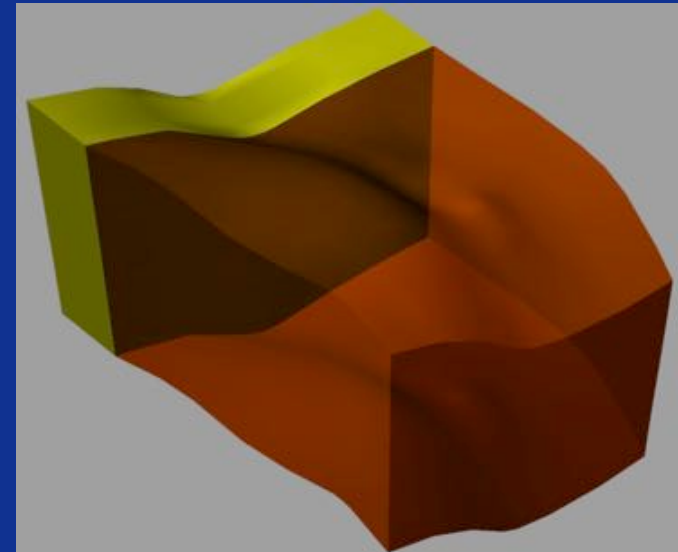
- Curves



- Surfaces



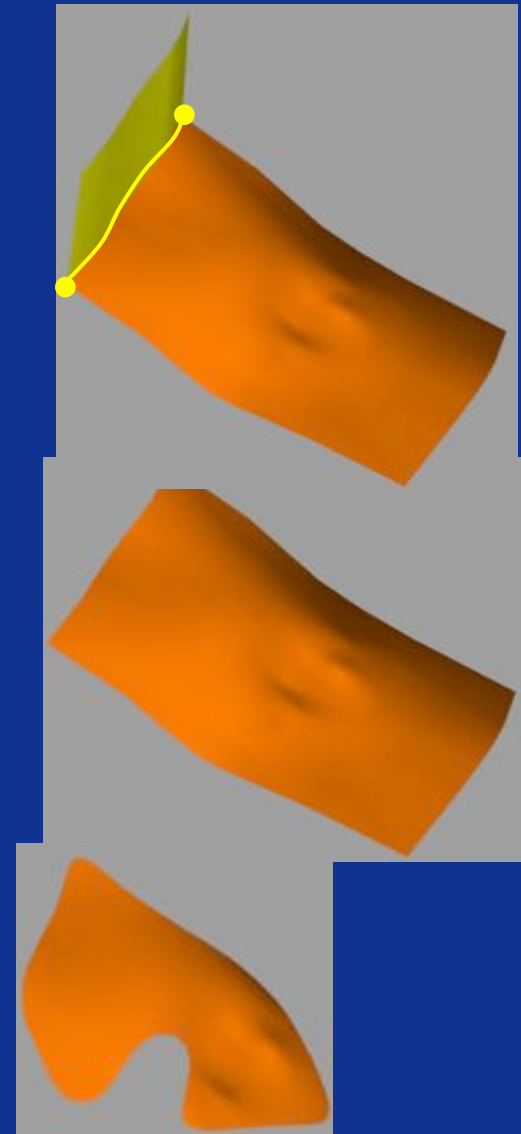
- Poly-Surfaces



= *analytic geometry*

Decisive features of “analytic” geometry

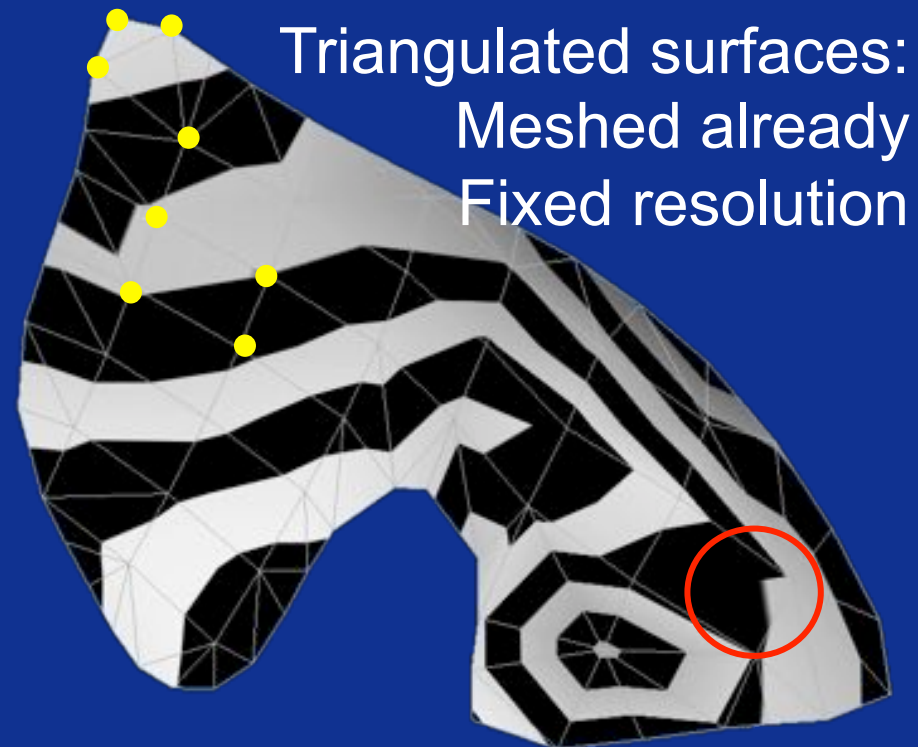
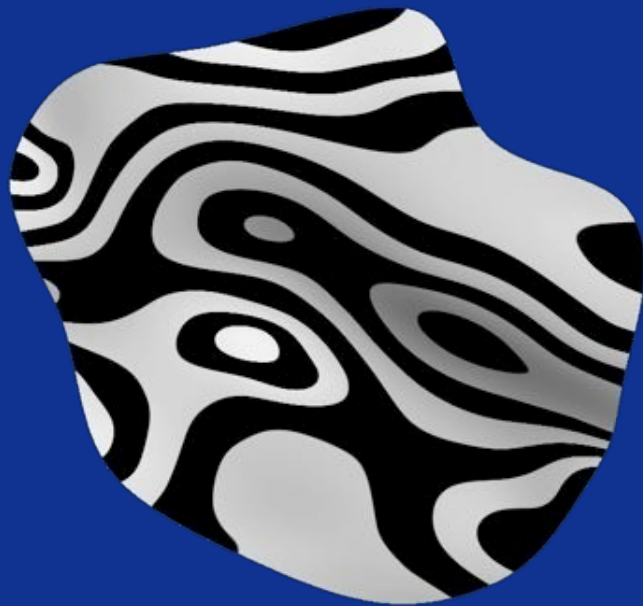
- To represent discontinuities in lines one require multiple *curves*
- Curves are interlinked by shared end *points*
- Analytically described surfaces are *smooth*
- To represent *discontinuities* in surfaces one requires multiple surfaces
- These *poly-surfaces* are stitched together with curves
- *Trimmed surfaces* are bounded by *trimming curves*
- Polysurfaces enclose *volumes*
- Such isolated volumes can be tagged by *material points*



Analytic geometry versus triangulated surfaces

Analytic geometry:

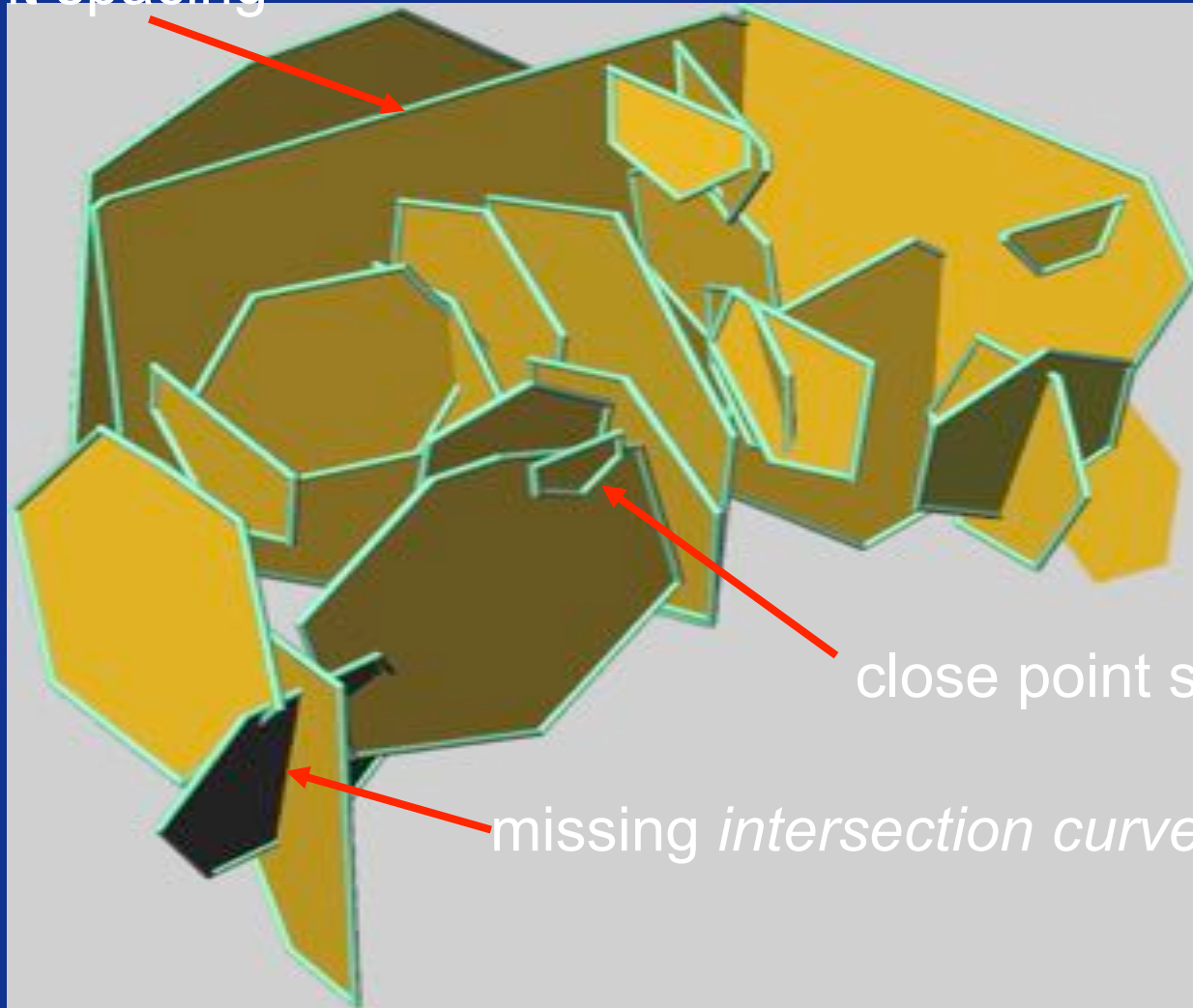
- Resolution independent
- Smooth / continuous
- Versatile



Triangulated surfaces:
Meshed already
Fixed resolution

Geometric entities also must know of each other

wide point spacing



close point spacing

missing *intersection curve*

Fracture polygons generated by FRED

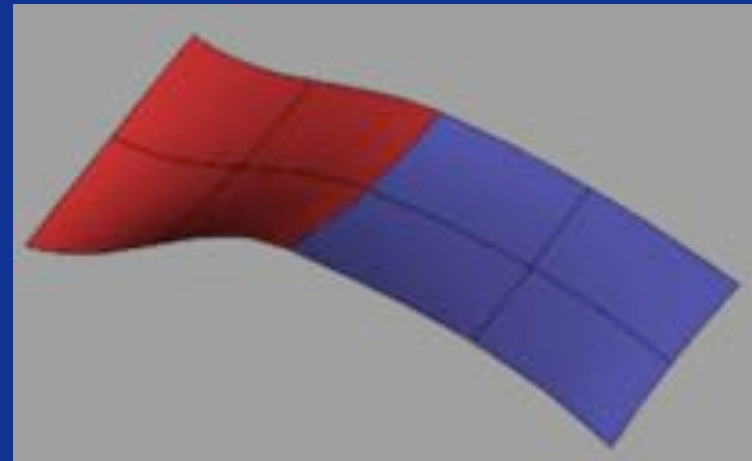
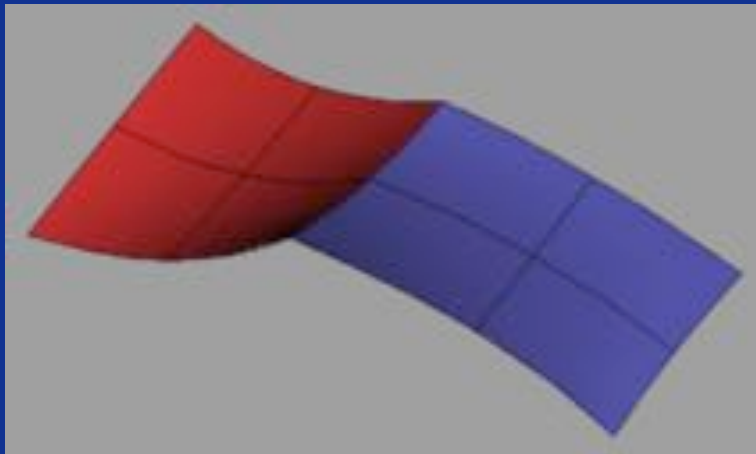
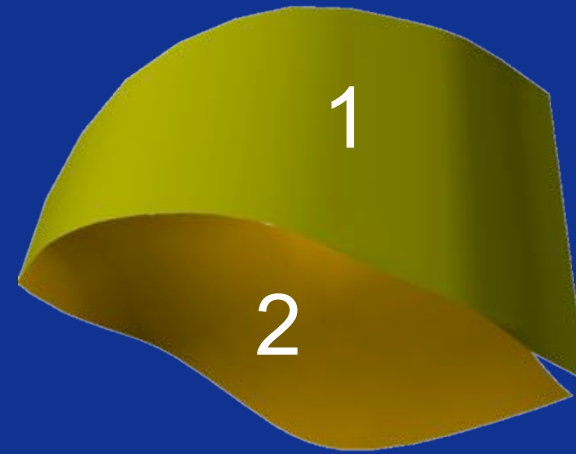
Surfaces divide volumes into sub regions

- In order to do this, subdividing surfaces = *internal boundaries* must be water tight
- Otherwise, “flood fill” operations starting from *material points* during meshing will transgress boundaries through gaps



Matching of surfaces

- By position
- By tangency
- By curvature

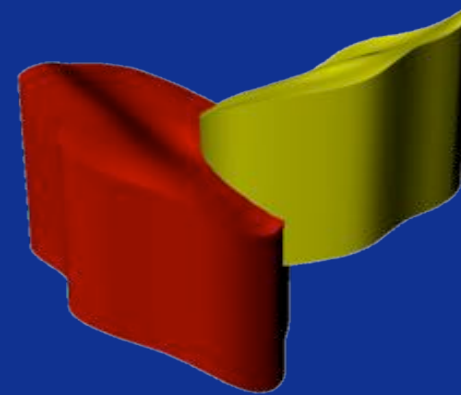


Model topology

- In surface-based models, like ours, volumetric entities consist of poly-surfaces which are shared among them
- Volumes are identified by *material points*
- The *model topology* clarifies the facing relationships between such entities (“lumps”) and their component surfaces, e.g., inside, outside etc.
- This topological tree is essential for boolean type operations with surface-based sub-volumes
- Boolean operations are *union, intersection, difference* etc.
- The boolean operations are essential for the representation of geological cross-cutting relationships

Boolean operations

Union



Original entities

Difference O_2



Intersection



Volumetric subdivisions must be unique !

Summary

- Consistent CAD is based on a hierarchy of *points*, *curves*, *surfaces* and *poly-surfaces*
- Individual entities (curves and surfaces) are defined by analytical functions with a continuous derivative
- The representation of discontinuities requires multiple curves or surfaces
- The poly-surfaces of a model yield a unique subdivision of its volume into *sub-volumes*
- Individual sub-volumes are identified by *material points* which must lie within them
- For this to work the sub-volumes must be “*water tight*”
- The surface entities have to know of each other; they must have been *intersected*.
- The geometric relationships of points, curves, and surfaces shared among objects are defined in terms of the *model topology*

Exercise

- Using *Rhinoceros*, generate some points in space
- Connect them with curves – change their order
- Join curves into loops
- Create surfaces from the curves with loft and patch commands
- Trim one of the surfaces with a curve
- Split surfaces using curves and other surfaces
- Match two surfaces by position, tangency and then by curvature
- Join the matching surfaces into a poly-surface
- Blend spatially separated surfaces
- Find intersection curves of intersecting surfaces
- Move each surface to a separate drawing layer
- *Use How-To-Rhino instructions to prepare model for meshing*

2. Reservoir simulation with CSMP

Model Configuration and Reservoir Simulation with CSMP Discrete Fracture Reservoir Simulator

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CSMP++ reservoir simulation workflow

1. Model building using BREP via NURBS or faceted representation
2. Meshing in ANSYS, quality check, output to CSMP and selection of model regions for simulation
3. Conversion of model into CSMP native binary format
4. Two-phase configuration
 - a. Assignment material properties, initial fluid properties and values
 - b. Essential conditions
 - c. Computing initial conditions
 - d. Time stepping strategy
 - e. Well schedule
 - f. Selection of output time steps
5. Property modelling (partially overwriting initial values)
6. Running the simulations
 - a. Model initialization
 - b. Checking screen output
 - c. Model output to files
 - d. Evaluating a (verbose) run over a few time steps
 - e. Running simulations under LINUX, protecting it with a screen
 - f. Trouble shooting
7. Visualisation of results using *Paraview*, *Excel* and *Maple*
8. Post-processing to find grid-block scale flow properties

Initial model checking

- Mesh diagnostics have to be passed
 - Disconnected vertices – delete them in Tetra
 - duplicate vertices, edges, elements - delete
 - Non-manifold vertices – visualise and fix, check for underlying geometrical / family problem
 - sense of node-numbering error (surface/volume orientations) – negative Jacobian (detected in element volume computation)
 - pits in outer wall of model (uncovered faces) – leads to pfvrt number error when VSet is checked
 - overlapping (2D or penetrating (3D) elements –fix
- Region names must be less-than or equal to 20 characters long!
- Over-constrained elements (e.g., triangles with 3-nodes, flow cannot be simultaneously parallel to two edges)
- Model is too big for current machine (2.1GB on 32bit PC, normally at more than 400 k nodes, 1.5 m elements)
- Steady-state pressure equation does not converge, see residual of SAMG solver - or – solver convergences to incorrect values

AMG workcount and statistics

=====

```
number of cycles :      11
L2-resid (before):  0.122D+09
L2-resid (after) :  0.727D-17
average reduction:  0.509D-02
average reduction:  0.723D-01 < 2 cycles skipped>
```

Analysis steps and options

- (0) Convert model in binary CSP files (CANNOT BE SKIPPED)
- (1) Run simulator with configuration read during (0)
- (2) Run simulator with some new configuration (same variables)
- (3) INJECTOR - PRODUCER set-up
- (4) Effective permeability calculation for block model (automatic k_{eff} , q_f/q_m analyzer)
- (5) Like (4) but with custom boundary setup specified by user
- (6) Transient pressure (well) testing without multiphase flow
- (7) Fault permeability modelling
- (8) Mapping of property point data to element properties in CSMP
- (x) Analyse results in Maple or Excel

Always watch a few output steps for:

- Convergence of the solver (L2 norms)
- Meaningful results

Different types of multiphase runs

Field scale with wells

- Always transient, but with or without (initial) steady-state (starting out with constant pressure works only when gravity effects are off)
- transient -> analysis works better when inflow = outflow

Block analysis

- constant pressure differential (Dirichlet b.c.)
- steady-state is mandatory for relative permeability analysis
- constant rate (Dirichlet + Neumann b.c.); Neumann condition is applied via finite-volume integration of fluxes over outflow face of model

Outline

1. Input files and selection of model subregions
2. Model configuration from file
3. Simulation and runtime parameters
4. Output files
5. Evaluation of trial runs
6. Longterm simulations under LINUX

Selection of parts of mesh via *-regions.txt file

*.asc file lists regions and element types:

```
fracs100
'fracs100/fracs100_1.asc' generated by ICEMCFD on 06-Oct-04
21 # Number of families
# Objectname      Elementtype      Material-ID      Number of elements
MATRIX            TETRA_4          0                246334
MATRIX            HEXA_8          0                19062
MATRIX            PYRA_5          0                43859
MATRIX            PENTA_6         0                344
FRACTURES         TRI_3           0                36632
FRACTURES         QUAD_4          0                2019
FRACTURES         BAR_2           0                4833
LEFT              TRI_3           0                1128
LEFT              QUAD_4          0                222
...
# now the elements which make up each object are listed in sequence
MATRIX TETRA_4 246334
0 1 2 3 4 5 6 7 8 9
...
```

region name: no more than 20 chars!



*-regions.txt file specifies regions for use in simulation:

```
'fracs100_1-regions.txt' file SKM 6/8/04
no properties
FRACTURES
MATRIX
```

*Warning: Excluding volumetric regions may
create stand-alone (disconnected mesh)*

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File overview

- A wide range of files constitute the “input deck” for the CSMP reservoir simulator.
- All file names start with the name of the current model, but have different extensions starting with “-”.
- This section explains their roles and structure for a model called **modi**:

ANSYS Mesh

modi.asc
modi.dat
modi.log

Region Selection & Configuration

modi-regions.txt
modi-well_completions.txt
modi-well_schedule.txt
modi-2phase_flow_domain.txt

CSMP binary

modi.vset
modi_regions.dat
modi_boundaries.dat
modi_variables.dat

Property Modeling

modi-fault_regions.txt
modi-porosity_to_points_map.txt

Placement of variables (*-variables.txt' file)

- Node variables, $p_f, s_w, s_n, \rho_w, \rho_n, \mu_w, \mu_n$
 - are defined and stored at the FEM nodes
 - are interpolated between nodes using the FEM interpolation functions N :
 - CSP reservoir simulator interpolation is piecewise linear
 - Nodes coincide with bary-centers of FV
- Element variables, v_t, k, ϕ
- Have a unique – constant value on each FEM
 - Vary discontinuously from element to element
- Composite variables (relperms etc.)
 - To model phase transport, node variables have to be interpolated to FV interfaces (facets)
 - Facet values are related in a non-linear fashion to nodal values

Properties specified in physical variables file

Example

CSP Variable Name	Notation	SI Unit	Type	Minimum	Maximum	FE-Placement
(fracture) aperture	a	m	1	1.E-06	1.E-01	element
(fracture) compressibility	z	Pa ⁻¹	1	1.E-13	1.E-04	element
fluid volume source	q	m ³	1	-1.E+00	1.E+00	element
nodal fluid volume source	nq	m ³	1	-1.E+00	1.E+00	node
fluid pressure	p_r	Pa	1	-1.E+09	1.E+09	node
previous fluid pressure	p_{r0}	Pa	1	-1.E+09	1.E+09	node
absolute fluid pressure	p_f	Pa	1	1.E+05	1.E+09	node
total system compressibility	c_t	Pa ⁻¹	1	1.E-15	1.E-05	element
permeability	K	m ²	3	1.E-25	1.E+00	element
porosity	ϕ	X	1	1.E-05	1.E+00	element
velocity	v_t	m s ⁻¹	2	-1.E+02	1.E+02	element
volume flux	F	m s ⁻¹	1	0.E+00	1.E+01	element
nodal volume flux	NF	m s ⁻¹	1	0.E+00	1.E+01	node
relative permeability model	RPM	NA	1	1	4	element
brooks corey parameter	λ	NA	1	1	3	element
van genuchten parameter	vg	NA	1	1	6	element
entry pressure	p_d	Pa	1	0.E+00	5.E+07	element
residual saturation wetting phase	sw_r	X	1	0.E+00	5.E-02	element
residual saturation non-wetting phase	sn_r	X	1	0.E+00	1.4E-01	element
density oil	ρ_w	kg m ⁻³	1	1.E+02	1.8E+03	node
density water	ρ_n	kg m ⁻³	1	5.E+02	1.E+03	node
relative fluid density	ρ	kg m ⁻³	1	-1.5E+03	1.5E+03	element
viscosity oil	μ_o	Pa s	1	1.E-05	1.E-01	node
viscosity water	μ_n	Pa s	1	1.E-04	1.6E-03	node
saturation oil	S, Ψ_n	X	1	-5.E-02	1.05E+00	node
previous saturation oil	Ψ_{n0}	X	1	-5.E-02	1.05E+00	node
saturation water	Ψ_w	X	1	-5.E-02	1.05E+00	node
relative permeability oil	kr_n	X	1	0	1	element
relative permeability water	kr_w	X	1	0	1	element
total mobility	λ_t	X	1	1.E-23	1.E+00	element
pore volume	V	m ³ m ⁻³	1	1.E-20	1.E+08	element

Current variables

name	MParameter	unit	index	mininum		maximum		place	
thickness		h	m	1	1.00E-06	1.00E+03	element		
permeability				K	m2	1	1.00E-25	1.00E+00	element
fracture permeability				KFR	m2	1	1.00E-25	1.00E+00	element
fault breccia permeability			KBR	m2	1	1.00E-25	1.00E+00	element	
fault zone permeability			KFZ	m2	6	1.00E-25	1.00E+00	element	
porosity				PHI	X	1	1.00E-05	1.00E+00	element
pore volume				PV	m s-1	1	1.00E-20	1.00E+08	element
fluid pressure				PF	Pa	1	-1.00E+09	1.00E+09	node
previous fluid pressure				PFP	Pa	1	-1.00E+09	1.00E+09	node
compressibility rock				CR	Pa-1	1	1.00E-15	1.00E-05	element
total system compressibility			CT	Pa-1	1	1.00E-15	1.00E-05	element	
previous fluid volume source			QVP	m3	1	-1.00E+00	1.00E+00	element	
fluid volume source				QV	m3	1	-1.00E+00	1.00E+00	element
nodal fluid volume source			NQV	m3	1	-1.00E+00	1.00E+00	node	
gravity term				G	Pa	2	-1.50E+08	1.50E+08	element
face gravity term				FG	Pa	2	-1.50E+08	1.50E+08	face
total mobility				LT	m s-1	1	1.00E-20	1.00	
element									
mobility oil				LO	m s-1	1	1.00E-20	1.00	
element									
mobility water				LW	m s-1	1	1.00E-20	1.00	
element									
face total mobility				FLT	m s-1	1	1.00E-20	1.00	
face									
capillary diffusivity				DC	m2 s-1	1	1.00E-50	1.00E+00	element
total velocity				VT	m s-1	2	-1.00E+02	1.00E+02	element
volume flux				PVF	m3 s-1	1	0.00E+00	1.00E+02	element
previous volume flux				VF	m3 s-1	1	0.00E+00	1.00E+02	element
relative permeability model			KRM	NA	1	0.0		5.0	
element									

The *-configuration.txt file

- File header
- Five data blocks separated by blank lines:
 1. Default material properties & initial conditions
 2. Region-specific property values
 3. Region-specific flags (DIRICH, NEUMANN etc.)
 4. Essential conditions
 5. Run settings and output time steps
- Comments preceded by # sign
- Columns are tab delimited

Example

'box25-configuration.txt' for single-phase flow analysis in 3D

← empty line

(1) global (default) material properties

permeability		1.0e-14
porosity	space	0.25
fluid volume	source	0.0

tab

(2) regional material properties (overwriting #1)

LAYER1	complete	permeability	1.0e-14
LAYER1	complete	porosity	1.0
WELL_DM1X	complete	fluid pressure	1.0e6

(3) regional variable flags

WELL_DM1X	perimeter	fluid pressure	DIRICH
-----------	-----------	----------------	--------

(4) essential conditions on model boundaries

BOUNDARY_LEFT	interior	Dirichlet	saturation water	0.1
BOUNDARY_LEFT	complete	Dirichlet	fluid pressure	1.0e6

5. computational settings

time stepping	conservative
duration	94608000.
output time	3600.
output time	86400.

Meaningful choice of input parameters

- fracture-matrix partitioning
 - $q_m > q_f$
 - $q_m \sim q_f$
 - $q_f > q_m$
- mobility ratio (μ_n/μ_w)
 - ~ 1
 - ~ 2
 - > 10
- gravity effects ($g = |dp_f/\rho g z|$)
 - $\Delta\rho = 200 \text{ kg m}^{-3}$
 - $v_g = v_t$
 - $v_g = 2 v_t$
 - $v_g = 5 v_t$
- capillary effects ($p_d \geq 100$)
 - water wet
 - mixed wet
 - oil wet

Adjustable material & fluid properties

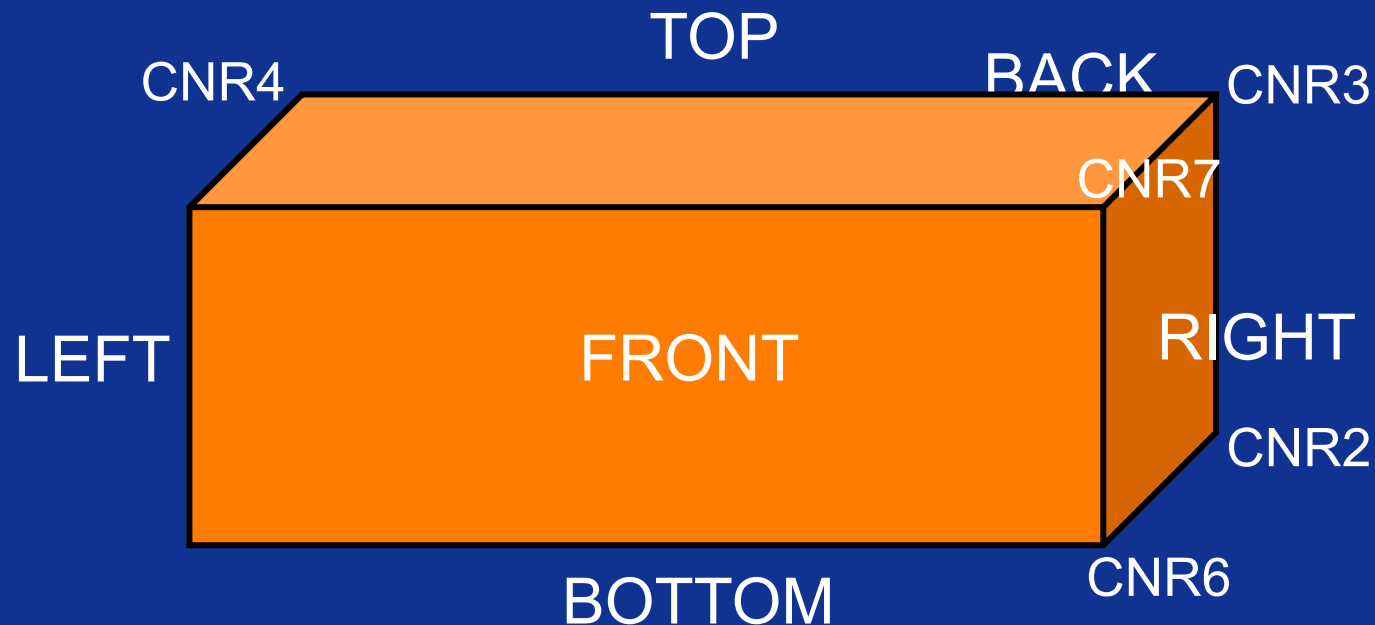
permeability	1.0e-12
porosity	0.25
density oil	800.0
density water	1000.0
viscosity oil	1.0e-3
viscosity water	1.0e-3
residual saturation water	0.03
residual saturation oil	0.14
total system compressibility	1.0e-9
Relative permeability model	
fracture	
rock matrix	
Rel. perm. model parameters	
entry pressure	10.
Initial conditions	
initial saturation oil	0.95
initial saturation water	0.05

Always check ranges in “-variables.txt” file and that these are honored when configuration is read.*

Essential conditions (Data block 4 in config. file)

- Dirichlet conditions can be applied anywhere to the model
- Overwrite any other conditions
- Are essential for a unique solution (can be replaced by initial conditions in transient case)

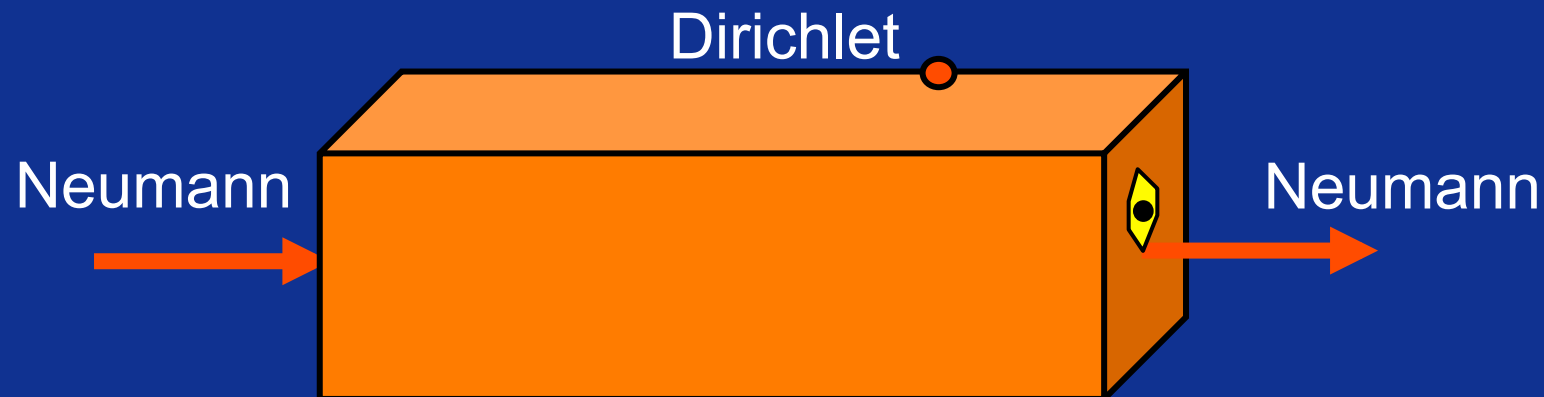
LEFT complete Dirichlet fluid pressure 1e7



Non-essential conditions

- 2nd order conditions / gradients / fluxes / aquifer models...
- Non-essential because unique solution can be obtained without them
- Imposed via boundaries: “face fluid volume source” and applied using surface integrals in PDE_Integrator
- NB: What goes in should match what goes out otherwise the run is over soon...

BOUNDARY_RIGHT complete Neumann -1e-6



Initial conditions

- Needed: initial values of saturation, fluid density, fluid viscosity, fluid pressure etc.
- Assigned in data blocks 2 & 3 of configuration file
- Initial “fluid pressure” is only taken into account in transient simulations (default) – runtime option *compute initial steady-state* must be deselected (option=0)
- Initial steady state is always computed when gravity is active. This happens automatically if the densities of water and oil are set to different values
- Saturation initialization (capillary equilibration) phase for model is possible but currently not accessible through file-based interface
- **Note: In CSMP variable values are initialized to flag:ANY, value:NAN, but Paraview cannot read datasets with NANs**

Well management: modi-well_schedule.txt

- Well-by-well tables with cumulative production in thousands of m3
- Incomplete (pressure data are missing etc.)
- CSMP computes rates from cumulative production

```
well_schedule_csmp.TXT # NAM RONG DOI MOI FIELD WELL SCHEDULE FROM JAN 2010 TO DEC 2012
```

```
DM2X
```

```
3493. 4130. # completion interval
```

```
# Reservoir condition total history
```

```
1.0946 1.133 # Boi, Bo
```

```
1.059629 1.059629 # Bwi, Bw
```

# Time (secs)	cum. liquids (km3)	cum. oil(km3)	cum. water (km3)	water cut (X)	GOR (km3/m3)
86400	0.71612398	0.71612398	0	0	28.6047
2678400	22.200002	22.200002	0	0	28.6047
5097600	41.2636334	41.2636334	0	0	28.6108

Which wells? – modi-well_completions.tx

- File header and
- Just a list of well names preceded by number, telling how many there are

`modi-wells-completions.text`

`10 DM2X DM3X DM406 DM408 DM410 R20 R25 R420 R421 R422`

Well management

- WellManager and Well classes
- Monitoring included for purpose of history matching

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Simulation steps (starting with CSMP native binary)

- Step 1: Read mesh and variables from *.vset & *.dat files into appropriate container objects.
- Step 2: Initialize PropertyDatabase
- Step 3: Build simulation model from mesh data, forming regions.
- Step 4: Assign material properties, initial conditions, and boundary conditions from *-configuration.txt file.
- Step 5: Assign relative permeability model and compute initial flow properties
- Step 6: Setup of monitoring of variables: for instance: "saturation oil" and "fluid pressure". VTU save of input variables.
- Step 7: Setup steady-state and-or transient fluid pressure computation with the SAMG solver. If computation of initial steady-state fluid pressure was requested, it is computed now starting with the boundary pressures, else user-specified initial fluid pressure is used. VTU save of initial pressure, velocity etc.
- Step 8: Explicit (setting: pedantic) or implicit (any other setting) finite volume advection scheme is setup. User chooses time stepping strategy. If "daring" is chosen, user is prompted for the PV fraction in which she/he is prepared to overstep CFL (in super-CFL elements, saturation functions are linearized)
- Step 10: Time-stepping loop is setup and initial time-stepping criteria are established.
- Step 11: Timestepping begins. In each time step the following operations are attempted:
 - Update time-increment and a check whether results need to be output in the current time interval.
 - Advance the saturation front by solving the transport equation.
 - Monitor property ranges and integrate saturation over model sub volumes.
 - Output variables to VTU files at user-specified output times.
 - Output model for restart
 - Re-compute two-phase flow properties, capillary pressure etc.
 - Re-compute fluid pressure if required.
 - Re-compute total velocity and capillary pressure gradients.
- When user-specified duration of the run has been reached the program terminates and the memory in use is returned to the operating system. The programme now outputs final saturations, runtime statistics etc

Runtime parameters

- Verbose (on=1, off=0), use 1 in trial run
- Initial steady state (compute=1; else 0)
- Choice of relative permeability model for rock matrix:
 - Brooks-Corey / Linear (if λ is set to 0)
 - Experimental
 - FractureMatrixUpscaled
- Linear model is invoked by setting Brooks-Corey parameter in configuration file to zero.

Duration and output time

- in seconds
- Guess duration using k_{eff} and the q_f/q_m ratio (look at upscaling slides for this), taking into account the pore volume which may need to be computed
- If constant rate computation is done inject 3-5 pore volumes but be consistent throughout sensitivity
- Get some quick output to verify that the run is O.K. (i.e., initial conditions etc.)
- Avoid too much output
- Do not make output intervals too large toward the end of run (only when output is written integral properties are computed)

Screen output

Read screen output during trial simulations because it will:

- Indicate potential problems with mesh
- Tell whether the variable values specified in your configuration file were accepted or reset (test against *-variables.txt file)
- Report whether SAMG solution converged
- Report simulation progress
- Give ranges of the constitutive variables that were computed
- Report any other issues

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Output

VTU files

model time in minutes



`modi-input_data.vtu`

permeability, porosity, total mobility, pressure, saturation oil

`modi-runtime_data60.vtu`

fluid pressure, velocity, volume flux for entire model

`modi_flow_domain720.vtu`

as previous, but only within flow domain.

Text based output from run

`modi-monitored-regions.txt`

'fluid pressure' and 'saturation oil' ranges as a function of time (secs) for each region in the model

`modi-saturation.txt`

'saturation oil' as a function of time (secs). This text file and data can be visualized directly in Excel or other spreadsheet application programs of choice.

`modi-simulation.txt`

parameter ranges at every timestep

`Runtime.log`

Warnings and error messages in chronological order.

Saved model

`modi-snapshot.vset` etc.

For restart of model or diagnosis what went wrong.

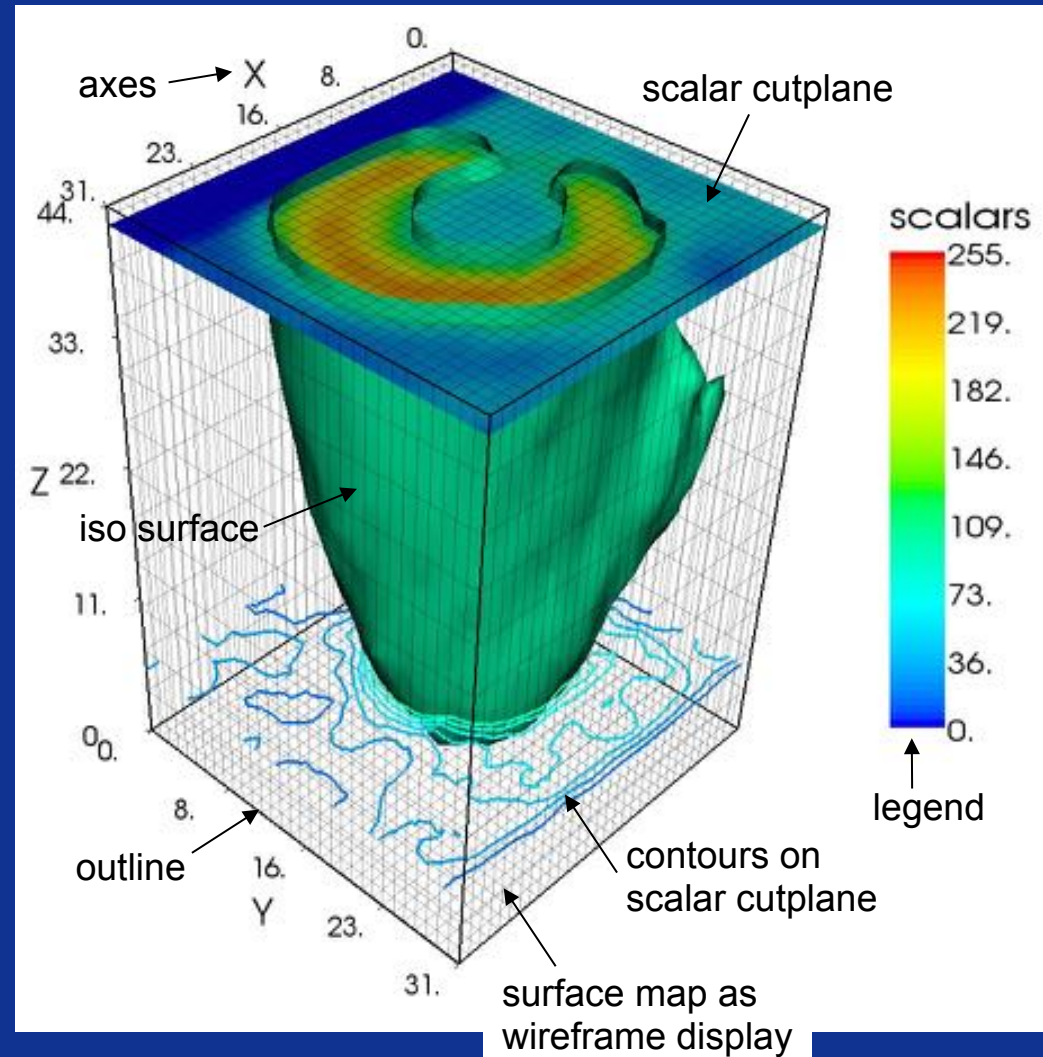
Text output for spreadsheets

'hex2_3-monitored-regions.txt' group data recorded up to time, t = 400000 seconds.					
groupname	volume	surface area			
IN_FRACTURE	3.50008	1000			
MATRIX	2498.75	5000			
Integrated values of properties:					
saturation oil					
time	IN_FRACTURE	MATRIX			
60	3.31774	2372.62			
1800	3.29388	2372.45			
...					
400000	0.496339	2340.23			
Value ranges of properties:					
fluid pressure					
time	IN_FRACTURE		MATRIX		
39.2776	1.00E+06	5.00E+06	1.00E+06	5.00E+06	
60	1.00E+06	5.00E+06	1.00E+06	5.00E+06	
...					
400000	1.00E+06	5.00E+06	1.00E+06	5.00E+06	
saturation oil					
time	IN_FRACTURE		MATRIX		
0	0	0.95	0	0.95	
...					
399995	0	0.60303	0	0.95	
400000	0	0.603029	0	0.95	

1. Review of *MayaVi* visualisation exercise

Types of displays:

- *Outline* (convex hull)
- *Axes*
 - Setting number formats (%6.3g)
- *Surface map*
 - contours
 - transparency
- *Legend*
 - color palette & lookup table
- *Scalar cut plane*
- *Iso-surface* – polydata normals



Outline

1. Input files and selection of model subregions
2. Model configuration from file
3. Simulation and runtime parameters
4. Output files
5. Evaluation of trial runs
6. Longterm simulations under LINUX

Memory requirements & runtime (2006)

Results obtained on 1.1 Ghz PIII laptop computer by running csp_res_sim.exe from the command line:

- 200 fracture model (200 x 30 x 200 m),
- mesh resolution 0.1-15 m (Ø FE-edge length 2 m),
- ~50,000 nodes, 120,000 elements,
- “Aggressive” gravity drainage run (density contrast between water and oil=200 kg m³)
- ~220 MB RAM peak memory usage
- runtime=1.5 days at water breakthrough
- Possible speedups
 - Optimization with Intel compiler (x 2-10)
 - Code efficiency tuning (20%)

Things that can go wrong

- Mesher may not be able to achieve target tolerances
- Model may be non-valid due to non-manifold vertices etc.
- Model may be too big for physical memory of computer
Win32 RAM limit is 2.1GB ~300 k DOF.
- Pressure or saturation SAMG solution may not converge
- Run may creep along if errors and warnings are raised
- Water injected on top of oil may also slow computation speed
- Assigned material property value may be outside of physically meaningful range
- Injection rate would initiate hydraulic fracture
- Total injection rate is unequal to total production rate

Outline

1. Input files and selection of model subregions
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Running under LINUX in screen environment

To protect long runs and put them into the background run them using the 'screen' environment under LINUX

```
screen ./res_sim.exe ...
```

- Use CTRL ^ a d to detach screen
- Paraview and MayaVi are also available under LINUX so you can visualize your results

On to the
code and
simulator...

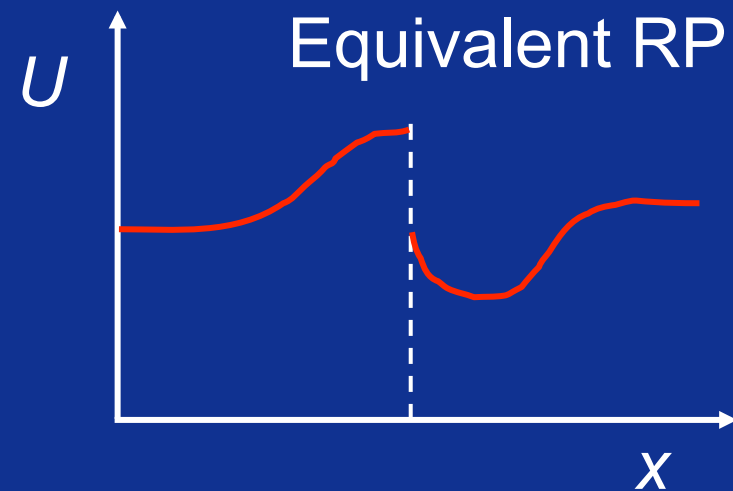
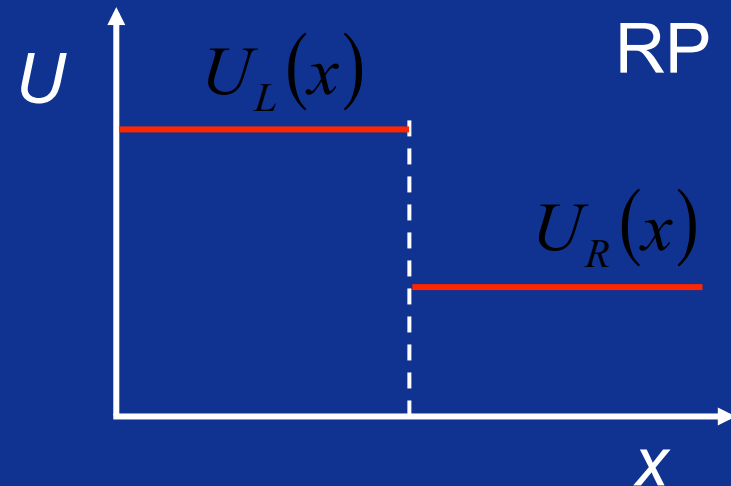
Some more theory...

The (generalized) Riemann problem (RP)

- What is the time-dependent solution of initially piecewise constant problem?

$$U(x, t = 0) = \begin{cases} U_L & \text{for } x \leq x_0 \\ U_R & \text{for } x > x_0 \end{cases}$$

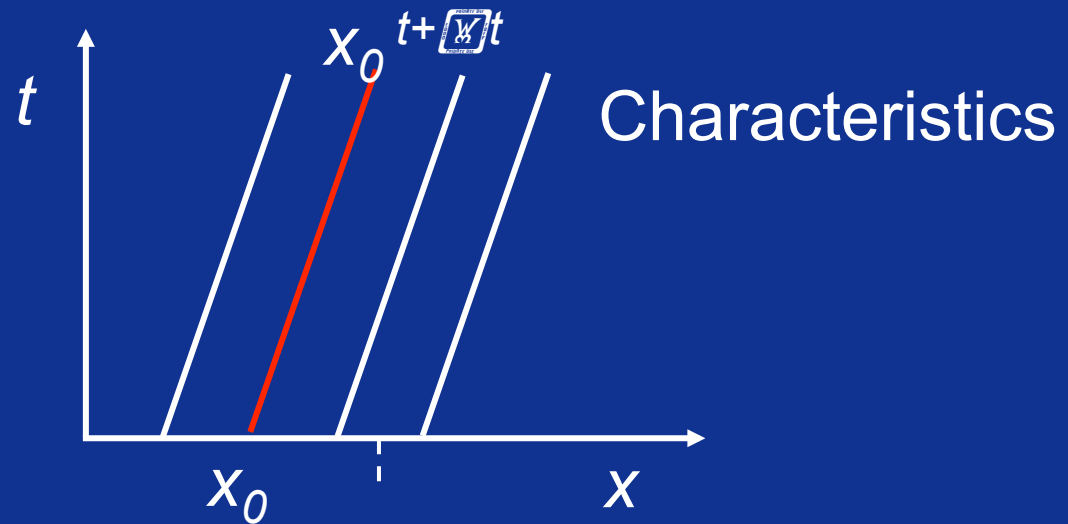
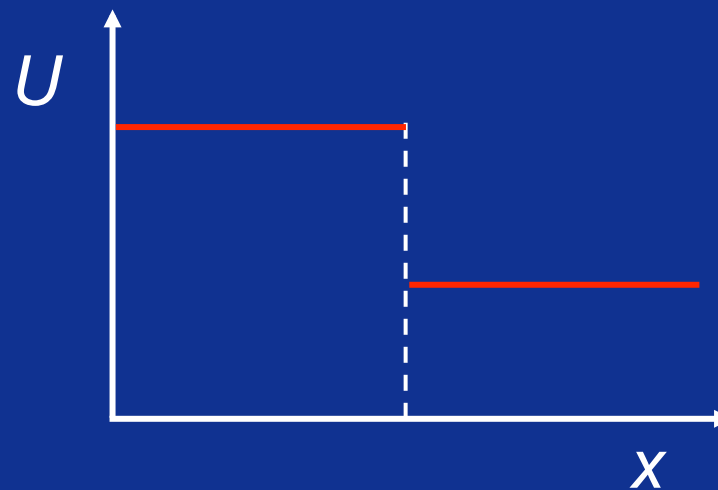
- Initial value problem
- Space extends indefinitely in positive and negative directions
- Gudonov schemes satisfy RP in integral sense



The Riemann problem in the time domain

Characteristics:

- Lines of time-invariant speed are used to visualize evolution of profile with time



The Buckley-Leverett equation

1. Shock wave

$$c_s = \frac{f(U_L) - f(U_R)}{U_L - U_R} \quad U(x_0, t) = \begin{cases} U_L & \text{if } c_s \geq x_0 \\ U_R & \text{if } c_s < 0 \end{cases}$$

2. Rarefaction wave

$$s_n(x) = \begin{cases} s_{nL} & \text{if } x - x_0 \leq \lambda_L \mathbf{v}_t t \\ s_n^* & \text{if } \lambda_L \mathbf{v}_t t \leq x - x_0 \leq \lambda_R \mathbf{v}_t t \\ s_{nR} & \text{if } x - x_0 \geq \lambda_R \mathbf{v}_t t \end{cases}$$

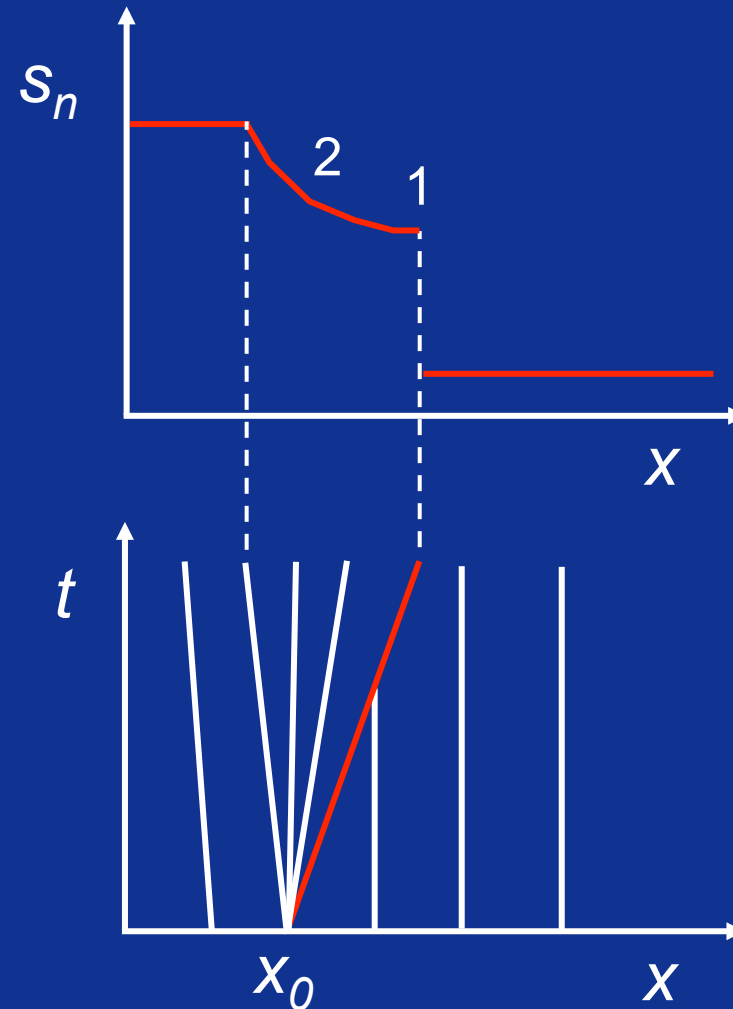
3. Mixed wave (at front)

$$s_n(x_0, t) = \begin{cases} s_{nL} & \text{if } \lambda_L > 0 \\ \lambda^{-1}(0) & \text{if } \lambda_L \leq 0 \leq c_s \\ s_{nR} & \text{if } c_s < 0 \end{cases}$$

- Iterative solution required

4. Contact discontinuities

$$\lambda = \frac{df}{dU} = \frac{df}{ds_n} (s_n) \mathbf{v}_t$$



Semi-discrete central upwind scheme for hyperbolic conservation laws

New method to estimate interface flux (and conserved variable value) at interface:

$$a_{\min} = \min(0, |\mathbf{v}_L|, |\mathbf{v}_R|)$$

$$a_{\max} = \max(0, |\mathbf{v}_L|, |\mathbf{v}_R|)$$

$$\Delta a = a_{\max} - a_{\min}$$

$$q_i = \frac{a_{\max} U_L - a_{\min} U_R}{\Delta a - [f(a_{\max}) - f(a_{\min})] / \Delta a}$$

adapted from Kurganov, Noelle, Petrova (2001), SIAM J. Sci. Comput. 23:3, 707-740.